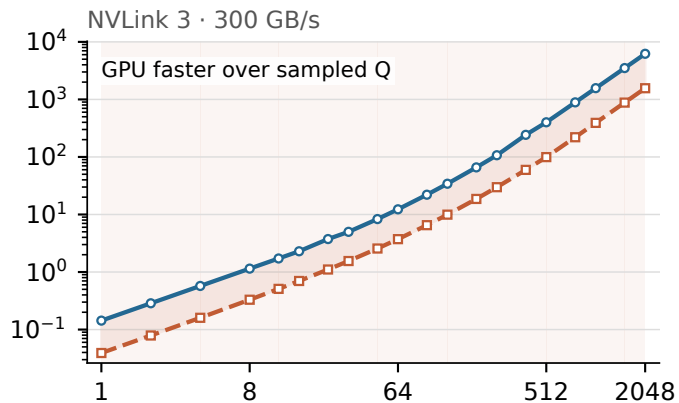


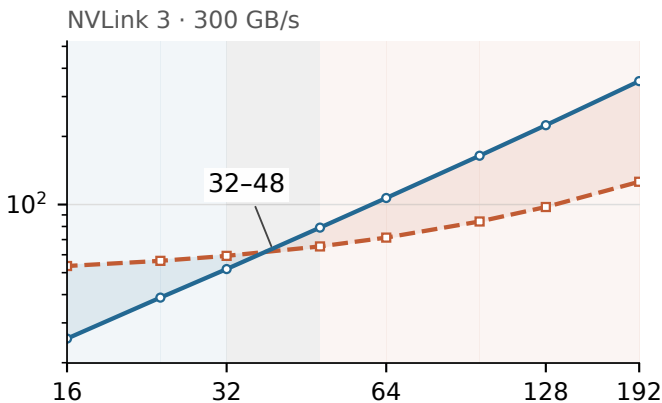
Prefill attention crossover · LLAMA-65B

GPU PIM PIM lower cost GPU lower cost

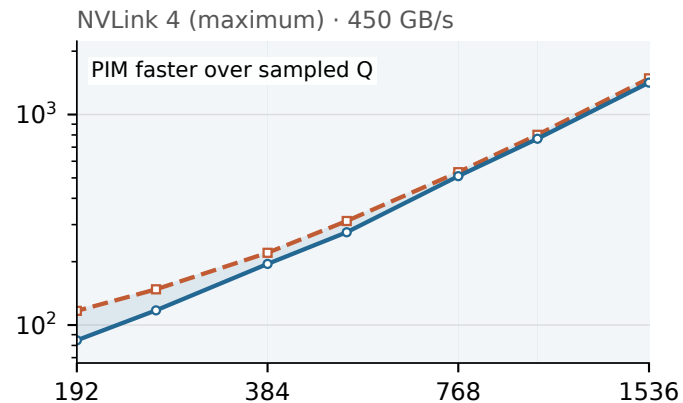
(a) Without MQ · Cache: 0



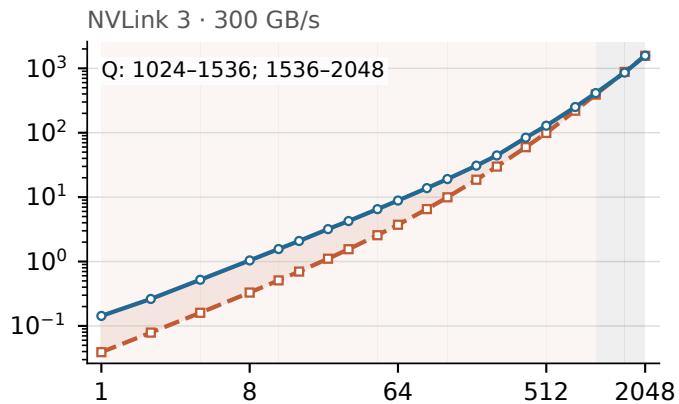
(b) Without MQ · Cache: 1,024



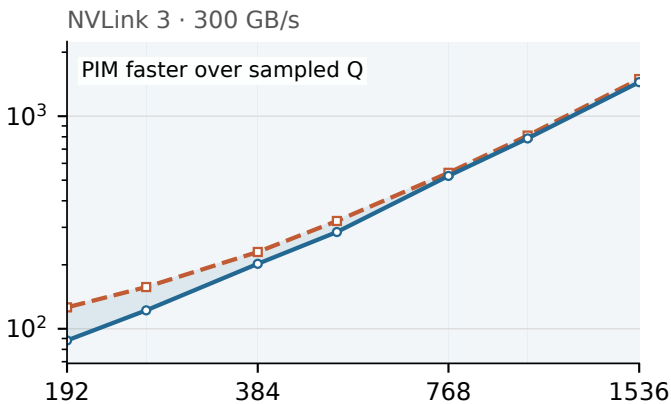
(c) With MQ · Cache: 1,024



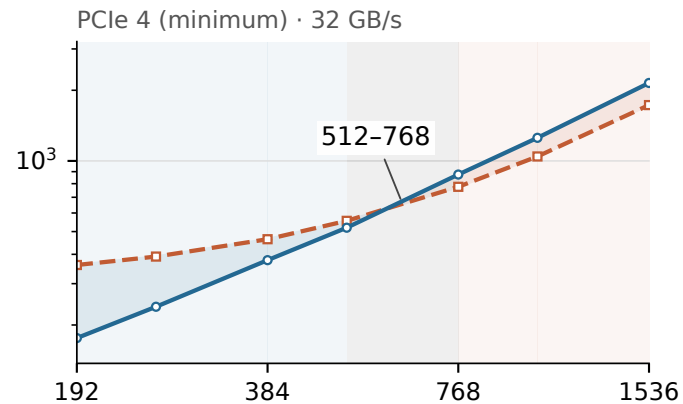
(d) With MQ · Cache: 0



(e) With MQ · Cache: 1,024



(f) With MQ · Cache: 1,024



New query tokens, Q